4-Channel 24-Bit Precision Data Acquisition System

STM32F407 + TI ADS1256 | bare-metal firmware, FreeRTOS, on-device DSP | evidence pack

What it is

A calibrated data-acquisition front end for ex vivo arterial haemodynamics: multi-catheter pressure and flow, time-synchronised with A-line ultrasound wall tracking, streaming live to a PC over USB. Written to the registers - no vendor HAL, no CMSIS pack, no dynamic allocation.

System

Parameter	Value
MCU / ADC	STM32F407VGT6 @ 168 MHz / TI ADS1256, 24-bit delta-sigma
Acquisition, muxed	4 differential pairs, 1875 complete scans/s, 133 us /DRDY deadline
Acquisition, wideband	1 channel, 30 kSPS, 33.3 us deadline, M=4 decimation on-device
Architecture	3-stage FreeRTOS pipeline, lock-free SPSC rings in CCM
Interface	hand-written USB OTG_FS device stack, CDC-ACM, CRC-16 framed binary protocol
Calibration	Q16.16 two-point, append-only flash log, 2048 slots, power-fail safe
On-device DSP	polyphase anti-alias decimator, 1024-pt Welch PSD, live RMS and THD+N
Memory	no heap linked; 40.3 KB flash, 41.3 KB SRAM, 26.8 KB CCM

Pipeline

```
/DRDY ISR (prio 5)  latch TIM2 + notify  ->  sampler (4) SPI
  -> ring_raw -> processor (3) decimate, calibrate, encode, CRC
  -> ring_tx  -> uplink (2) batch 16 packets -> USB bulk IN
  -> ring_dsp -> metrics (1) RMS / FFT / Welch / THD+N -> ring_meta -> uplink
```

Priorities are rate-monotonic, by deadline. A USB stall costs ring occupancy, not samples: the 256-slot tx ring absorbs 137 ms of host misbehaviour before anything is lost.

Verified without hardware

Everything in this section is reproducible from the repository on any machine, in under a minute. Bench-measured figures are reported separately and only once measured.

Check	Result
Firmware build (arm-none-eabi, -O2)	clean at -Wall -Wextra -Wconversion -Wsign-conversion -Wshadow -Wcast-align
Unit tests, portable core	6 CTest suites, 100 % pass, under ASan + UBSan
Unit tests, host tooling	43 pytest, 100 % pass, Python 3.9 / 3.11 / 3.12
Line coverage, portable core	95.1 % (468/492 lines), 93.5 % of functions
Static analysis	cppcheck warning/style/performance/portability: clean
Generated-code drift	CI regenerates filter coefficients and fails on diff
Repository	14 commits, 94 files, 10,145 lines of C / Python / asm

Test coverage, by module

The native tests compile the *same* source files the firmware links - `firmware/lib/*.c` - not a host reimplementation. A bug fixed by these tests is fixed in the image.

Module	Lines	Covered	Rate	What it is
`crc.c`	17	17	100.0 %	CRC-16/CCITT-FALSE, CRC-32/ISO-HDLC
`fft.c`	48	48	100.0 %	radix-2 FFT, Hann window
`fir_decimate.c`	39	39	100.0 %	polyphase decimator
`packet.c`	152	146	96.1 %	wire codec, framing, resync
`metrics.c`	110	103	93.6 %	Welch PSD, RMS, THD+N
`calibration.c`	51	47	92.2 %	Q16.16 two-point correction
`spsc_ring.c`	75	68	90.7 %	lock-free ring buffer
`TOTAL`	492	468	95.1 %	

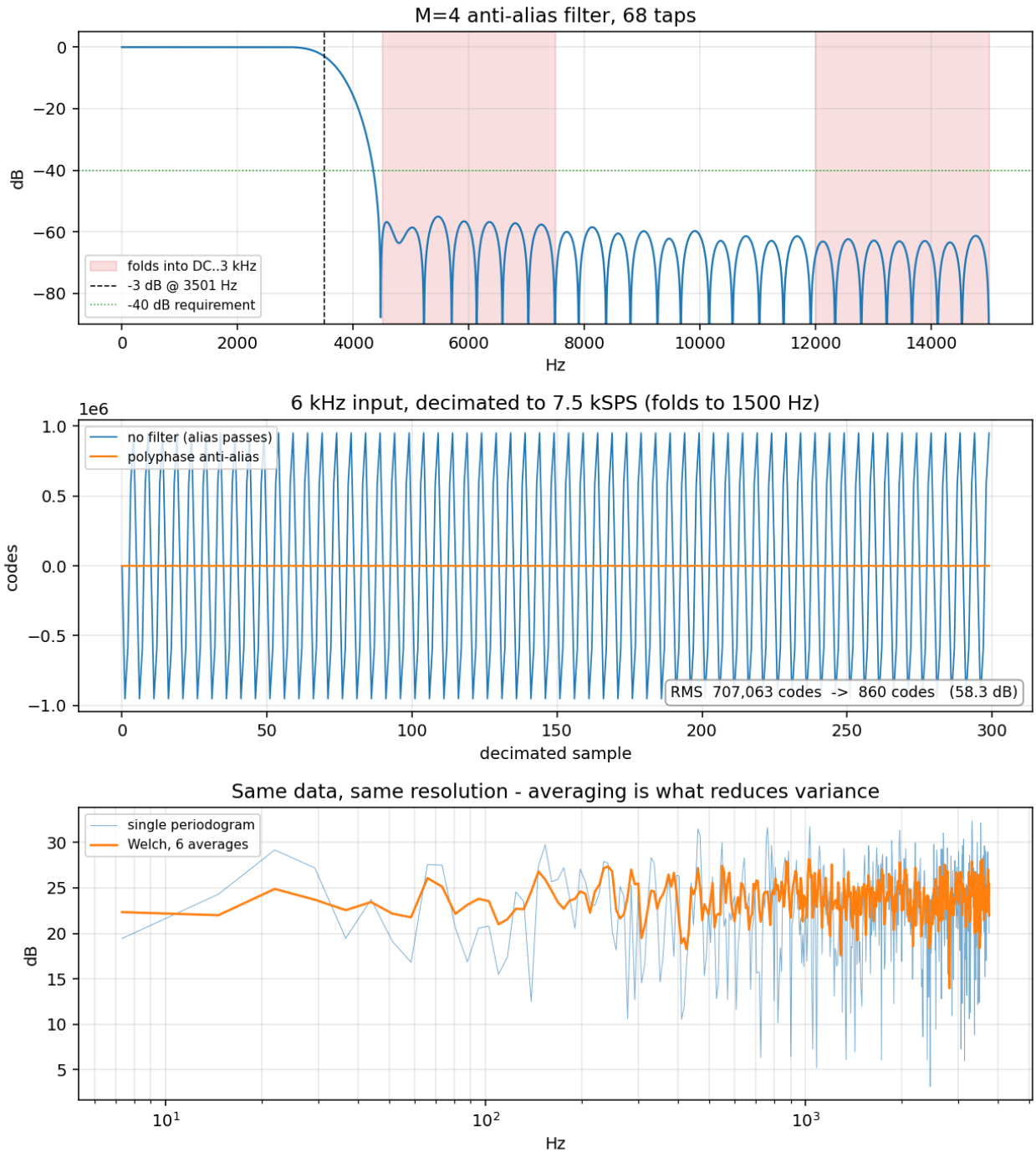
Cases the tests actually cover

- **Ring index wrapping past 2^32** - happens after ~6 days of continuous streaming and is otherwise untestable on target.
- **Exhaustive single-bit corruption** of every wire frame, all 176 bit positions, each one rejected.
- **Framing against a payload containing the sync word** - a receiver that trusts 0xA5 0x5A alone locks onto sample data within minutes.
- **Polyphase output bit-identical to naive filter-then-discard** - the optimisation is only worth claiming if it is provably free.
- **A control that fails**: the same out-of-band tone through an unfiltered decimator, rejected by 0.0 dB.
- **Fixed-point saturation rather than wraparound** near full scale, where int32 truncation would silently flip a sign.

Signal processing - computed and test-verified

Property	Value
Decimation	30000 Hz -> 7500 Hz, M = 4, 68 taps (17 per branch)
-3 dB corner	3501 Hz
Passband droop, DC-3 kHz	0.06 dB
Worst-case alias rejection	55.6 dB into DC-3 kHz (requirement: > 40 dB)
Measured on a 6 kHz tone	58.3 dB rejected | no filter (control): 0.0 dB
MAC rate	510 kMAC/s polyphase vs 2.04 MMAC/s naive - exactly 4x
Fixed-point fidelity	max 0.499 codes vs float with the same taps (half-LSB rounding only)
PSD variance, 1024-pt x1	5.37 dB bin scatter
PSD variance, 16384-pt x1	5.56 dB - 16x the resolution, no improvement
PSD variance, 1024-pt x6	2.00 dB - averaging is what reduces variance
THD+N, 1 kHz + 5 % H3	-25.3 dB (expected -26 dB)
THD+N, shorted input	reported INVALID - no tone, so no number

DSP verification plot



Top: the M=4 anti-alias response, with the input bands that fold into DC-3 kHz shaded and the 40 dB requirement marked. Middle: a 6 kHz tone decimated to 7.5 kSPS - it folds to 1500 Hz and arrives at full amplitude without the filter. Bottom: one periodogram against the 6-average Welch estimate, same data, same resolution. Regenerate with `ads1256 dsp --plot`.

Reproducing all of the above

```
cmake -B build -DCMAKE_TOOLCHAIN_FILE=cmake/arm-none-eabi-gcc.cmake
cmake --build build        # firmware, warning-free

cmake -B build-host -DDAQ_BUILD_HOST_TESTS=ON
ctest --test-dir build-host  # 6 suites, ASan + UBSan

pip install -e host[dev]
cd host && python -m pytest -q  # 43 tests

ads1256 dsp              # DSP chain, no hardware needed
```